Tamilnadu - Grade 10

2015 - Mathematics

Maximum Marks: 100

1. If $f(x) = x^2 + 5$ then $f(-4) =$ __________.

(a) 26

(b) 21

(c) 20

(d) –20

Answer: (b) 21

Solution: $(-4)^2 + 5$

2. If $1 + 2 + 3 + \ldots\ldots + n = k$, then $1^3 + 2^3 + \ldots.. + n^3$ is equal to:

(a) k^2

(b) k^3

(c) $\frac{k(k+1)}{2}$

(d) $(k + 1)^3$

Answer: (a)

Solution: $k^2 = 21$

3. If the n^{th} term of a sequence is $100n + 10$, then the sequence is _____.

(a) an A.P.

(b) a G.P.

(c) a constant sequence

(d) neither A.P. nor G.P.

Answer: (a)

Given n^{th} term = 100n + 10

This can be rewritten as 110 + (n – 1) 100

This is in form of $T_n = a + (n - 1) d$ which forms an A.P.

So, option (a) is correct.

4. The remainder when $x^2 - 2x + 7$ is divided by $x + 4$ is:

(a) 28

(b) 29

(c) 30

(d) 31

Answer: (d) 31

Solution: $x + f\sqrt{x^2 - 2x + y}\left(x - 6\right)$

$$\frac{x^2 \pm 4_x}{0 - 6\cancel{x} + 7}$$

$$\frac{\underset{+}{-}\cancel{6x}\underset{+}{-}24}{31}$$

5. Let, b = a + c. If the equation $ax^2 + bx + c = 0$ has equal roots, then:

(a) a = c

(b) a = –c

(c) a = 2c

(d) a = 2c

Answer: (a) a = c

Solution: $b^2 - \Delta ac = 0$

$(a + c)^2 - 4ac = 0$

$a^2 + c^2 + 2ac - 4ac = 0$

$a^2 + c^2 - 2ac = 0$

$(a - c)2 = 0$

$a = c$

6. $(5 \; x \; 1)\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = (20)$ then the value of x is:

(a) 7

(b) –7

(c) $\frac{1}{7}$

(d) 0

Answer: (b) –7

Solution: 5 × 2 – x + 3 = 20

x = 13 – 20 = –7

7. The straight line $4x + 3y - 12 = 0$ intersects the y – axis at:

(a) (3, 0)

(b) (0, 4)

(c) (3, 4)

(d) (0, –4)

Answer: (b) (0, 4)

Solution:

8. The value of k, if the straight lines $3x + 6y + 7 = 0$ and $2x + ky = 5$ are perpendicular is:

(a) 1

(b) –1

(c) 2

(d) $\frac{1}{2}$

Answer: (b) –1

Solution:

9. ΔABC is a right angled triangle where $\angle B = 90°$ and $BD \perp AC$. If BD = 8 cm, AD = 4 cm, CD is:

(a) 24 cm

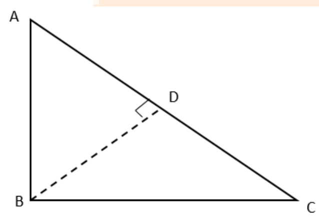
(b) 16 cm

(c) 32 cm

(d) 8 cm

Answer: (a) 24 cm

Solution: Given BD = 8 cm, AD = 4 cm, $\angle ABC = 90^o$ and $BD \perp AC$



We know that, if a perpendicular is drawn from vertex containing a right angle of a right triangle then, triangle formed on each side is similar to each other.

So, Δ DBA and Δ DCB are

Similar,

$$\frac{BD}{CD} = \frac{AD}{BD}$$

$$BD^2 = AD \times CD$$

$$8^2 = 4 \times x$$

$$x = \frac{64}{4} = 16\,cm$$

So, option (b) is correct.

10. Triangles ABC and DEF are similar. If their areas are 100 cm^2 and 49 cm^2 respectively and BC is 8.2 cm, then EF is ______.

(a) 5.47 cm

(b) 5.74 cm

(c) 6.47 cm

(d) 6.74 cm

Answer: (b) 5.74 cm

Solution:

11. If $\tan\theta = \frac{a}{x}$ then the value of $\frac{x}{\sqrt{a^2+x^2}} =$

(a) cos θ

(b) sin θ

(c) cosec θ

(d) sec θ

Answer: (a) cos θ

Solution:

12. $(\cos^2\theta - 1)(\cot^2\theta + 1) + 1 =$

(a) 1

(b) –1

(c) 2

(d) 0

Answer: (d) 0

Solution: $\left(\cos^2\theta - 1\right) = \frac{\left(\cos^2\theta + \sin^2\theta\right)}{\sin^2\theta} + 1$

$$-\frac{\left(1-\cancel{\cos^2}\theta\right)}{1-\cancel{\cos^2}\theta}+1$$

$-1 + 1 = 0$

13. The total surface area of a solid right circular cylinder whose radius is half of its height h is equal to _____.

(a) $\frac{3}{2}\pi\text{h sq. units}$

(b) $\frac{2}{3}\pi h^2$ sq. units

(c) $\frac{3}{2}\pi h^2$ sq. units

(d) $\frac{2}{3}\pi h$ sq. units

Answer: (c)

Solution: $r = \frac{h}{2}$

TSA = $2\pi rh + 2\pi r^2$

$$= 2\pi\left(\frac{h^2}{2} + \frac{h^2}{4}\right)$$

$$= 2\pi \times \frac{3}{4}h^2$$

$$= \frac{3}{2}\pi h^2$$

14. For any collection of n items $\left(\sum x\right) = \bar{x} =$ ______.

(a) $n\bar{x}$

(b) $(n-2)\bar{x}$

(c) $(n-1)\bar{x}$

(d) 0

Answer: (c)

Solution:

$\because\ \bar{x} = \frac{\sum x}{n}$

$\sum x = \bar{x}.n$

$So, \left(\sum x\right) = \bar{x}$

$\bar{x}.n = \bar{x}$

$\bar{x}(n-1)$

15. The probability that a non – leap year will have 53 Sundays and 53 Mondays is ______.

(a) $\frac{1}{7}$

(b) $\frac{2}{7}$

(c) $\frac{3}{7}$

(d) 0

Answer: (d)

16. Let P = {a, b, c}, Q = {g, h, x, y} and R = {a, e, f, s} then find:

(i) $P \setminus R$ and

(ii) $Q \cap R$

Solution: P / R = {b, c}

$Q \cap R = \{\ \}$

17. Let A = {1, 2, 3, 4, 5}, B = N and $f : A \to B$ be defined by $f(x) = x^2$. Find the range of f. identify the type of function.

Solution:

f(1) = 1

f(2) = 4

f(3) = 9

f(4) = 16

f(5) = 25

Range of f = {1, 4, 9, 16, 25}

Function is a square function.

18. Find the sum of first 10 terms of the series $1^2 - 2^2 + 3^2 - 4^2 +$

Solution: 1 – 4 + 9 – 16 ….. upto 10 terms

– 3 + (–7) + (–11) + ….. upto 10 terms

$$s_{10} = \frac{n}{2}\left[2a + (n-1)a\right]$$

$$= \frac{10}{2}\left[2\times1-3++(10-1)(-4)\right]$$

= 5[–6 –36]

= –225

19. Find the L.C.M. of 3(a – 1), $2(a - 1)^2$, $(a^2 - 1)$.

Solution: 3(a – 1) = 3 × (a – 1)

$2(a - 1)^2$ = 2 × (a – 1) (a – 1)

$(a^2 - 1)$ = (a + 1) (a – 1)

(cm = 3 × 2 × (a – 1) (a – 1) (a + 1)

= $6(a - 1)^2$ (a + 1)

20. Find the value of k for which the roots are real and equal in $12x^2 + 4kx + 3 = 0$.

Solution: a = 12 b = 4k c = 3

Δ = 0

$b^2 = 4_{ac\,=\,0}$

$16k^2 - 4 \times 12 \times 3 = 0$

$16k^2 = 12 \times 12$

$$k^2 = \frac{\cancel{12}\times\cancel{12}}{\cancel{16}} 3\times3$$

$$\boxed{k=3}$$

21. If $A = \begin{pmatrix} 2 & 3 \\ -9 & 5 \end{pmatrix} - \begin{pmatrix} 1 & 5 \\ 7 & -1 \end{pmatrix}$ then find the additive inverse of A.

Solution: $A = \begin{bmatrix} 2 & 3 \\ -9 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 5 \\ 7 & -1 \end{bmatrix}$

$= \begin{bmatrix} 1 & -2 \\ -16 & 6 \end{bmatrix}$

Additive inverse of $A = \begin{bmatrix} -1 & 2 \\ 16 & -6 \end{bmatrix}$

22. If the area of the ΔABC is 68 sq. units and the vertices are A(6, 7), B(–4, 1) and C(a, –9) taken in order, find the value of ‘a’.

Solution: Area $= \frac{1}{2}\left[x_1\left(y_2 - y_3\right) + x_2\left(y_3 - y_1\right) + x_3\left(y_1 - y_2\right)\right]$

$2 \times 6p = \left[6(1+9) + -4(-9-7) + a(7-1)\right]$

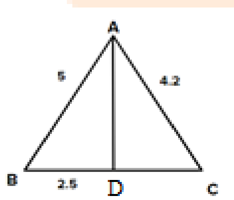
$13_6 = 60 + 64 + 6a$

6a = 136 – 124

$= \frac{\cancel{12}}{\cancel{6}} \boxed{a = 2}$

23. In ΔABC, the internal bisector Ad of ∠A meets the side BC at D. If BD = 2.5 cm, AB = 5 cm and AC = 4.2 cm, then find DC.

Solution:



In Δ ABD and Δ ACD

∠ BAD = ∠ CAD ⊂ Bisectors

AD = AD (common)

ΔABD ~ ΔACD

So, $\frac{BD}{CD} = \frac{AB}{AC}$

$\frac{2.5}{CD} = \frac{5}{4.2}$

$CD = \frac{\overset{0.5}{\cancel{2.5}} \times 4.2}{\cancel{5}}$

$= 2.10$ cm

24. Simplify $\cos\theta\sqrt{1+\tan^2\theta} + \sin\theta\sqrt{1+\cot^{2\theta}}$

Solution: $\cos\theta\sqrt{1+tan^2\theta} + \sin\theta\sqrt{1+\cos^2\theta}$

$\cos\theta \times \sec\theta + \sin\theta \times \text{cosec}\theta$

$(\because \ \sec^2\theta - \tan^2\theta = 1$

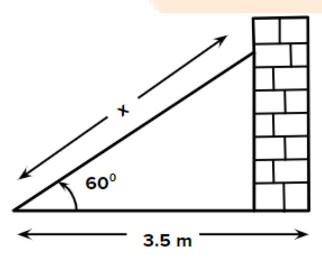
$\text{cosec}^2\theta - \cot^2\theta = 1)$

$\cancel{\cos}\theta \times \frac{1}{\cancel{\cos}\theta} + \sin\cancel{\theta} \times \frac{1}{\sin\cancel{\theta}}$

$1 + 1 = 2$

25. A ladder leaning against a vertical wall makes an angle of 60° with the ground. The foot of the ladder is 3.5 m away from the wall. Find the length of the ladder.

Solution:



Let length of ladder = x

$\cos 60 = \frac{3.5}{x}$

$x = 3.5 \times 2$

$= 7$ m

26. The radii of two right circular cylinders are in the ratio of 3 : 2 and their heights are in the ratio 5 : 3. Find the ratio of their curved surface areas.

$$\frac{R_1}{R_2} = \frac{3}{2} \qquad \frac{H_1}{H_2} = \frac{5}{3}$$

$$\frac{CSA_1}{CSA_2} = \frac{\cancel{2}\pi\, R_1 H_1}{\cancel{2}\pi\, R_2 H_2} = \frac{\cancel{3}}{2} \times \frac{5}{\cancel{3}}$$

$= 5 : 2$

27. Find the volume of a sphere-shaped metallic shot-put having diameter of 8.4 cm.

Solution: Volume = $\frac{4}{3}\pi r^3$

$$= \frac{4}{3}\pi \times (4 \cdot 2)^3$$

$= 310 \cdot 33$ cm^3

28. Calculate the standard deviation of the first 13 natural numbers.

Solution: To find $\bar{x}$

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{1+2+3+4+5+6+7+8+9+10+11+12+13}{13}$$

$$= \frac{\cancel{91}}{\cancel{13}} = 7$$

x	$x = \bar{x}$	$(x = \bar{x})^2$
1	–6	36
2	–5	25
3	–4	16
4	–3	9
5	–2	4
6	–1	1

7–A	0	0
8	1	1
9	2	4
10	3	9
11	4	16
12	5	25
13 $N = 13$	6	36 182

$$SD = \sqrt{\frac{\sum\left(x-\bar{x}i\right)^2}{N}} = \sqrt{\frac{18.2}{13}} = 3.74$$

29. When two coins are tossed together, what is the probability of getting at most one head?

Solution: n(c) = {HH, HT, TH, TT}

n(H) = {HT, TH, TT}

$$\mathrm{P}\left(\mathrm{H}\right) = \frac{3}{4}$$

30. (a) Determine whether the matrix product tis defined or not. If the product is defined state the dimension of the product matrix for $A_{2\times 5}$ and $B_{5\times 4}$.

(b) Find the equation of the straight line parallel to the line 3x – y + 7 = 0 and passing through the point (1, –2).

Solution: You can only multiply two matrices if and only if the number of columns is first equals to the no. of rows in see and matrix

Hence, matrix is undefined.

Product order is → $(AB)_{2\times 4}$

y = 3α + 7

m = 3

$\therefore$ for parallel $m_1 = m_2 = 3$

Equation of line. $y - y_1 = m(x - x_1)$ @ $(1_1 - 2)$

y + 2 = 3(x – 1)

y + 2 = 3x – 3

$\boxed{y = 3x - 5}$

31. U = {–2, –1, 0, 1, 2, 3} A = {–2, 2, 3, 4, 5} and B = {1, 3, 5, 8, 9}. Verify De Morgan’s law of complementation.

Solution de Margon’s low states that A \ (B ∩ C) = (A \ B) ∪ (A \ C)

LHS – ∪ \ (A ∩ B)

{–2, –1, 0, 1, 2, 3} \ (3, 5)

= {3}

RHS – (∪ \ A) ∪ (∪ \ B)

{–2, 2, 3} ∪ {1, 3}

= {–2, 2, 3, 1}

LHS ≠ RHS Hence defined.

32. Let A = {5, 6, 7, 8} B = {–11, 4, 7, –10, –7, –9, –13} and *f* = {(*x*, *y*); *y* = 3 – 2*x*, *x* ϵ A, *y* ϵ B}

(i) Write down the elements of *f*.

(ii) What is the co-domain?

(iii) What is the range?

(iv) Identify the type of function.

Solution:

y = 3 – 2x

y = 3 – 2 × 5 = - 7

y = 3 – 2 × 6 = - 9

y = 3 – 2 × 7 = - 11

y = 3 – 2 × 8 = - 13

(i) f : (5, -7) (6, -9) (7, -11) (8, -13)

(ii) B = {–11, 4, 7, –10, –7, –9, –13}

(iii) {-7,-9,-11,-13}

(iv) one-one and into

33. Find the sum of all 3 digit natural numbers which are divisible by 9.

Solution: series becomes 108, 117, …………… 999

a = 108

d = 9

$a_n = 999$

$a_n = a + (n - 1)d$

999 = 108 + (n – 1)9

$(n - 1) = \frac{999-108}{9}$

$= \frac{891}{9}$

n = 99 + 1

= 100

34. If m – nx + 28x^2 + 12x^3 + 9x^4 is a perfect square, then find the values of m and n.

Solution:

$$\left(ax^2+bx+c\right)^2 = a^2x^4+b^2x^2+c^2+2abx^3+2bcx+2cax^2$$

comparing we get

$$a^2 = 9 \Rightarrow a = 3$$
$$2ab = 12 \Rightarrow b = 2$$
$$b^2 + 2ca = 28 \Rightarrow c = 4$$
$$m = c^2 \Rightarrow m = 16$$
$$n = -2bc \Rightarrow n = -16$$

35. Sum of two numbers is 24 and sum of their reciprocals is $\frac{1}{6}$, find the numbers.

Solution: x + y = 24   y = 24 – x

$$\frac{1}{x}+\frac{1}{y}=\frac{1}{6}$$

$6(y + x) = xy$

$$6\left(24 - \cancel{x} + \cancel{x}\right) = x\left(24 - x\right)$$

$144 = 24x - x^2$

$x^2 - 24x + 144 = 0$

$x(x - 12) - 12(x - 12) = 0$

$x = 12, 12 \qquad y = 24 - x$

$= 24 - 12$

$= 12$

36. $A = \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix}$ and B = (13 –6) then verify that $(AB)^T = B^T A^T$.

Solution: $A = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \ A^T = \begin{bmatrix} -2 & 4 & 5 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 3 & -6 \end{bmatrix} \ B^T = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix}$

$\text{LHS} = (AB)^T = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 4 & 3 & -6 \end{bmatrix}$

$= [-2 + 12 + 30] = -20$

$B^T A^T = [-2 + 12 - 30] = -20$

37. Find the equation of the straight line passing through the point of intersection of the lines $5x - 8y + 23 = 0$ and $7x + 6y - 71 = 0$ and is perpendicular to the line joining the points (5, 1) and (–2, 2).

Solution: $5x - 8y = -23 \times 6$

$7x + 6y = 71 \times 8$

$$\rightarrow 30x - 4\cancel{8}y = -138$$

$$\begin{array}{r} 56x + 4\cancel{8}y = 568 \\ \hline 86x = 430 \end{array}$$

$$x = \frac{\cancel{430}}{\cancel{86}}5$$

8y = 5x + 23

$$y = \frac{\cancel{48}}{\cancel{8}}6$$

Point $\rightarrow$ (5, 6)

Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{2-1}{-2-5} = \dfrac{1}{-7}$

$$m_1 \times \frac{\cancel{-}1}{7} = \cancel{-}1$$

$$\boxed{m_1 = 7}$$

Equation of line pairing by (5, 6) and with $m_1 = 7$ slope

$y - 6 = 7(x - 5)$

$y - 6 = 7x - 35$

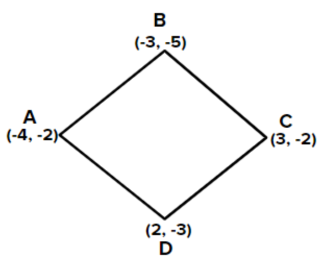
$7x - y = 29$

38. Find the area of the quadrilateral formed by the points (–4, –2), (–3, –5), (3, –2) and (2, 3).

Solution: Ar. Of ABC =

$$\frac{1}{2}\{-4(-5+2)+(-3)(-2+2)+3(-2+5)\}$$

$$= \frac{1}{2}\{12+0+9\} = \frac{21}{2}\ \text{cm}^2$$



$$\text{Ar. of CDE} = \frac{1}{2}\{3(3+2)+2(-2+2)+(-4)(-2-3)\}$$

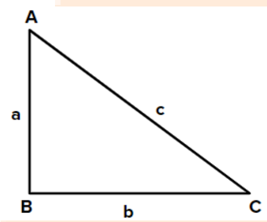
$$= \frac{1}{2}\{15+0+20\} = \frac{35}{2}$$

$$\text{Total are} = \frac{21}{2} \times \frac{35}{2} = \frac{56}{2} = 28\ \text{cm}^2$$

39. State and prove Pythagoras theorem.

Solution: Pythagoras theorem states that the perpendicular square plus the base square is equal to hypotenuse square.

$a^2 + b^2 = c^2$



40. From the top of a tower of height 60 m the angles of depression of the top and the bottom of a building are observed to be 30° and 60° respectively. Find the height of the building.

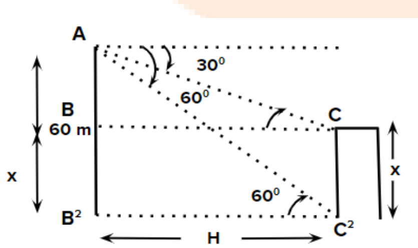
Solution: Let CE = x

DE = H

$\tan 30 = \frac{AB}{BC}$

$\tan 60 = \frac{AD}{ED}$

BC = ED from figure



$\underline{AB}\sqrt{3} = \frac{AD}{\sqrt{3}}$

(60 – x) × 3 = 60

180 – 3x = 60

120 = 3x

$\boxed{x = 40}$

41. The outer and inner radii of a hollow sphere are 12 cm and 10 cm respectively. Find its volume.

Solution: volume of hallow sphere $= \frac{4}{3}\pi\, r_1^3 - \frac{4}{3}\pi\, r_2^3$

$= \frac{4}{3}\pi\, 12^3 - \frac{4}{3}\pi \times 10^3$

$= \frac{4}{3} \times \frac{22}{7}(1728 - 1000)$

$= \frac{4}{3} \times \frac{22}{7} \times 728$

$= 3050.6\ cm^3$.

42. A spherical solid of radius 18 cm is melted and recast into three small solid spherical spheres of different sizes. If the radii of two spheres are 2 cm and 12 cm. Find the radius of the third sphere.

Solution: Vol. of Bigger sphere = vol. of 1 + vol. of 2 + vol. of 3

$\cancel{\frac{4}{3}} \times 18^3 = \cancel{\frac{4}{3}} \times \pi \times 2^3 + \cancel{\frac{4}{3}}\pi \times 12^3 + \cancel{\frac{4}{3}}\pi \times r^3$

5832 = 8 + 1728 + r^3

$\sqrt[3]{5832 - 1736 = r}$

r = 16 cm

43. Calculate the standard deviation of the following: 38, 40, 34, 31, 28, 26, 34.

x	**y**	**d**	**d^2**	$x - \bar{x}$	$(x - \bar{x})^2$

38	26	–8	64	5	25
40	28	–6	36	7	49
34	31	–3	9	1	11
31	34 – A	0	0	–2	4
28	34	0	0	–5	25
26	38	4	16	–7	49
34	40	6	36	1	1
$\bar{x}=33$					$\frac{1}{154}$

$$\sigma=\sqrt{\frac{\left(x-\bar{x}^2\right)}{n}}$$

$$=\sqrt{\frac{154}{7}}=\sqrt{22}=4.69$$

44. A bag contains 10 white, 5 black, 3 green and 2 red balls. One ball is drawn at random. Find the probability that the ball is white or black or green.

Solution: n(s) = 20

P(W/B/G) = 10 + 5 + 3

= 18

P(W/B/G) = $\frac{18}{20}$

$= \frac{9}{10}$

45. (a) Find the total area of 14 squares whose sides are 11 cm, 12 cm, … 24 cm respectively.

(b) If one root of the equation 2x2 – ax + 64 = 0 is twice the other, then find the value of 'a'.

Solution:

(a) To find total area of 14 square we apply the formula for sum of n natural no. which is

$\frac{n(n+1)(2n+1)}{6}$ which gives sum of squares.

sum of 14 squares = Total sum of 24 – Total sum of 1 to10

$$=\frac{24\times25\times49}{6}-\frac{10\times11\times21}{6}$$

= 4900 – 385

= 4515 cm^2

(b) Let roots α and 2 α

$$\alpha + 2\alpha = \frac{-(-a)}{2} \qquad \alpha + 2\alpha = \frac{64}{2}$$

$$4 + 8 = \frac{a}{2} \qquad 2\alpha^2 = 32$$

$$a = \boxed{24} \qquad \alpha^2 = 16$$

$$\alpha = 4$$

So, a = 24

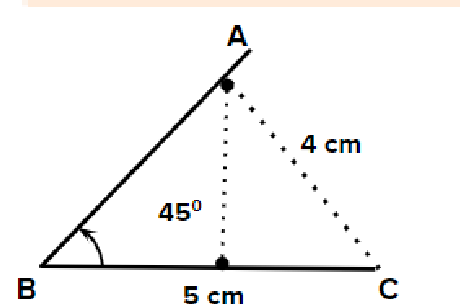
46. (a) Construct a ΔABC such that BC = 5 cm, ∠A = 45° and the median from A to BC is 4 cm.

OR

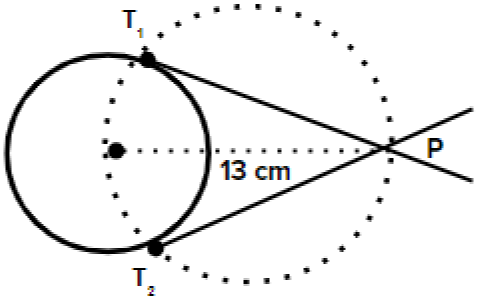
(b) Draw a circle of diameter 10 cm. From a point P, 13 cm away from its centre, draw the two tangents PA and PB to the circle and measure their lengths.

Solution:

(a) first draw BC of 5 cm. then from B draw line making 45° BC then take mid point of BC then from mid point of 4.5 that interest draw B making angle 45° Intersection more as A.



(b) Draw circle of sum radius Take point P at 13 cm from centre draw circle with mid of BCM name intersection as T_1 and T_2 measure length.



47.

(a) Solve graphically $2x^2 + x - 6 = 0$

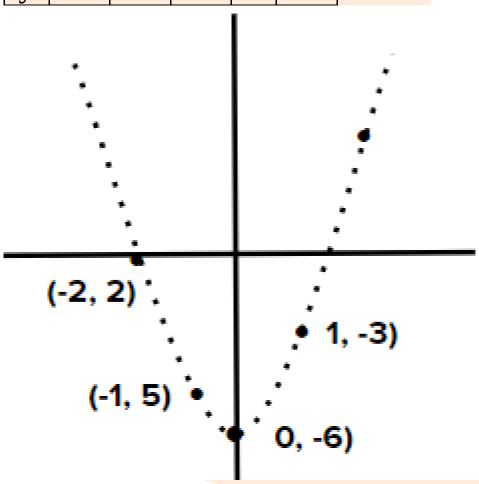
(b) Draw the graph of $xy = 20$, x, $y > 0$.

Use the graph to find y, when $x = 5$ and to find x when $y = 100$.

Solution:

(a) = y $2a^2$ + x – 6

x	0	1	–1	2	–2
y	–6	–3	–5	4	0



(b) xy = 20 $\Rightarrow y = \dfrac{20}{x}$

x	5	10	20	–5	–10
y	4	2	1	–4	–2

$@\, x = 5$

$\boxed{y = 4}$

$@\ y = 10$

$\boxed{x = 2}$